Neil Maushard

(309) 643-8455 | Peoria, IL | maushardn05@gmail.com | www.linkedin.com/in/neil-maushard

EDUCATION

University of Illinois at Urbana-Champaign (UIUC), Grainger College of Engineering — **Expected Dec 2026**
Bachelor of Science in Mechanical Engineering, Minor in Electrical Engineering — GPA: 4.0/4.0
Related Coursework:
Robot Dynamics and Controls, Intro Robotics, Mechatronics, Mechanical Design 1 & 2, Dynamics of Mechanical Systems, Electronic Circuits, Materials, Solid Mechanics, Statics, Fluid Mechanics

TECHNICAL SKILLS

Design Software: SolidWorks, Fusion360, Creo Parametric, AutoCAD, Altium, KiCAD
Fabrication & Prototyping: 3D Printing, Laser Cutting, CNC Machining, MIG Welding
Programming: Python, ROS, MATLAB, C++/C, OpenCV (Computer Vision)
Electronics/Controls: Simulink, Arduino, ESP32, Motors, Motor Controllers, IMUs, Encoders, Soldering, PCBs

PROJECT HIGHLIGHTS

Self-Balancing Robot — Dec 2025 – Present
- Designed and built a 2-wheeled self-balancing mobile robot with an ESP32, an IMU, DC motors, and a motor controller
- Implemented and tuned PID control using accelerometer and gyro data for stable balance and fast disturbance recovery
- Designed and iterated upon modular wheel and chassis system for easy assembly, disassembly, and access to internals

1-DOF 6-Legged Walking-Dispensing Robot — Sep 2025 – Dec 2025
- Designed and built a robot that walks 6 m/min while dispensing a payload every 2 m powered by a single DC motor
- Performed simulations of 4-bar linkage leg designs to refine foot path geometry and achieve desired walking gait
- Designed spur gear powertrain that transfers DC motor input to desired walking speed and dispensing frequency
- Analyzed walking-motion kinematics and dynamics to smooth acceleration profiles and lower transient force spikes

Drone Swarm Control System — Nov 2025 – Dec 2025
- Developed a decentralized potential field control algorithm with goal attraction and obstacle/inter-agent repulsion
- Implemented conditional sub-functions like a repulsive curl to handle edge cases like local minima and deadlocks
- Tuned and benchmarked control parameters to reduce average convergence time while maintaining 100% success rate
- Simulated hundreds of configurations, varying # of agents and goal/start/obstacle positions to evaluate robustness

6-Axis Robot Arm, *Electrical Team Lead*, American Society of Mechanical Engineers — Sep 2024 – April 2025
- Led 7 engineers to implement electronic systems into a 6-DOF robotic arm for UIUC's Engineering Open House
- Built Arduino-based stepper motor control system converting inverse kinematics inputs into 0.1°-accurate positioning
- Oversaw development of a closed-loop PID DC motor control system using motor controller, encoder, and Arduino
- Designed full robotic system in Altium and implemented it, integrating 20+ components and 60+ ft of wiring
- Collaborated with 5 mechanical team members to design links and joints in Fusion360 that securely house electronics
- Worked with 3 programming team members to debug control and pathing algorithms of electromechanical system

EXPERIENCE

Hendrickson, *Manufacturing Engineering Intern* — June 2025 – Aug 2025
- Eliminated need for inefficient part-marking tool by integrating part-marking into pre-existing laser cutting programs
- Characterized springback of a novel material relative to aluminum and steel, improving tolerances and reducing scrap
- Designed and welded a sliding-rail die rack to replace die stacking, improving ergonomics and speeding up tool swaps
- Designed tooling (vacuum, magnetic, chain-driven) to improve ergonomics and efficiency in handling heavy parts
- Collaborated with machine operators to gather firsthand insights, turning their feedback into actionable improvements
- Completed projects such as designing and integrating test fixtures, machinery components, and tool improvements

Hutchens Research Group, *Research Assistant*, UIUC — Jan 2024 – May 2025
- Designed new needle insertion experimental setup using SolidWorks, improving precision of data collection
- Optimized geometry of experimental setup to stay within deformation limits using SolidWorks FEA simulations
- Manufactured custom components for experimental setups using 3D printing, laser cutting, and CNC machining

- Tested new setup beyond operating conditions, noting modes of failure and iterating design to improve
- Analyzed and visualized data using Python and MATLAB for clear discussion of results with supervisors